STATE BOARD OF TECHNICAL EDUCATION, BIHAR

LABORATORY

PRACTICAL FILE

STRENGTH OF MATERIAL FOR CIVIL ENGG.

Course Code: 2415303 (P2415303)

Diploma in Civil Engineering • Semester III

Student's Name	
Enrollment No.	
Roll No.	
Branch	Civil Engineering
Semester	III
Session	20____ – 20____
Name of Institute	

Department of Civil Engineering

(Name & address of the institute)

How to write this file

Is file ke har practical ko nimn likhit format mein likha gaya hai. Apne college ke lab manual se practical ki sankhya aur readings confirm karke observation table mein apni actual readings bharein — yahan di gayi readings sample hain, sirf samajhne ke liye.

- **Aim** — practical ka uddeshya ek line mein.
- **Apparatus** — naam aur specification (capacity, IS code).
- **Theory / Formula** — principle aur formula jo calculation mein use hua.
- **Procedure** — step-wise, past tense mein, chhote vakya.
- **Observation table** — readings, units ke saath; IS code ka reference.
- **Calculation** — formula lagakar steps dikhayein.
- **Result** — answer with unit aur IS permissible limit ke saath.
- **Precautions** — 3 se 4 points.

General instructions

- Har practical ke baad date aur teacher ke signature zaroor lein.
- Diagram pencil se, scale se banayein; likhai blue/black pen se.
- Index page mein page number bharna na bhoolein — file check ke waqt yahi dekha jata hai.
- Practical exam (ELA) mein viva hota hai — aim, formula aur result yaad rakhein.

Index

Ye file SBTE Bihar curriculum P2415303 ke 13 practicals par aadharit hai. Page number index mein bharein aur har practical par teacher ka signature lein.

S. No.	Title of the Practical	Page No.	Date	Signature
01	Determination of moment of inertia of a flywheel and comparison with its theoretical value	4		
02	Study of different components of a Universal Testing Machine (UTM) and their functions	5		
03	Tension test on mild steel bar as per IS: 1608	6		
04	Compression test on a mild steel specimen using Universal Testing Machine	7		
05	Flexural test on a timber beam in both orientations as per IS: 2408	8		
06	Shear strength test on a given specimen	9		
07	Izod impact test on a given specimen	10		
08	Charpy impact test on a metal specimen	11		
09	Brinell hardness test on a metal specimen	12		
10	Rockwell hardness test on a metal specimen	13		
11	Preparation of models of different types of trusses	14		
12	Shear force and bending moment diagrams for a simply supported beam under point and distributed loads	15		
13	Determination of principal stresses and their plane using Mohr's circle	16		

PRACTICAL NO. 01

Determination of moment of inertia of a flywheel and comparison with its theoretical value

Aim

To determine the moment of inertia of a given flywheel experimentally and to compare it with the theoretical value calculated from its mass and dimensions.

Apparatus / Equipment Required

Flywheel mounted on ball bearings with axle and pointer; Set of weights, strong thread, stopwatch, vernier caliper, steel scale, weighing balance.

Theory

The moment of inertia of a body about an axis is the sum of the products of the masses of its particles and the squares of their distances from that axis, $I = \Sigma mr^2$; it measures the resistance of the body to angular acceleration. In this experiment a weight m hangs from a thread wound round the axle of radius r, and when it falls through a height h in time t its potential energy is converted into the kinetic energy of the weight and the rotational kinetic energy of the flywheel, part of the energy being lost in friction.

Formulae Used

- Velocity of the falling weight, $v = 2h / t$
- Angular velocity of the flywheel, $\omega = v / r$
- Loss of potential energy = $m g h$; K.E. of weight = $\tfrac{1}{2} m v^2$; rotational K.E. of flywheel = $\tfrac{1}{2} I \omega^2$
- $I = [\, 2 m g h - m v^2 \,] / \omega^2$ (friction neglected)
- Theoretical value for a disc, $I = \tfrac{1}{2} M R^2$

Procedure

1. The mass M and radius R of the flywheel, and the radius r of the axle, were measured.
2. A thread carrying a weight of 2 kg was wound round the axle, the weight being kept at a known height h.
3. The weight was released and the time t taken to reach the ground was measured with a stopwatch.
4. The experiment was repeated five times with the same weight and the mean time was taken.
5. The velocity v, angular velocity ω and the moment of inertia I were calculated.
6. The experimental value was compared with the theoretical value $\tfrac{1}{2}MR^2$.

Observation

Trial	Height h (m)	Time t (s)	v = 2h/t (m/s)	ω = v/r (rad/s)
1	1.200	8.10	0.296	11.85
2	1.200	8.05	0.298	11.92
3	1.200	8.00	0.300	12.00
Mean	1.200	8.05	0.298	11.92

Mass of falling weight m = 2.0 kg; radius of axle r = 0.025 m; mass of flywheel M = 25.0 kg; radius R = 0.160 m. Take g = 9.81 m/s². Readings are sample values.

Calculation

$v = 2 \times 1.200 / 8.05 = 0.298$ m/s ; $\omega = 0.298 / 0.025 =$ **11.92 rad/s**

P.E. lost = $m g h = 23.54$ J; K.E. of weight = $\tfrac{1}{2} \times 2.0 \times 0.298^2 = 0.089$ J; hence rotational K.E. = $23.54 - 0.089 = 23.45$ J = $\tfrac{1}{2} I \omega^2$

$I = (2 \times 23.45) / (11.92)^2 = 46.90 / 142.09 =$ **0.330 kg·m²**

Theoretical $I = \tfrac{1}{2} M R^2 = 0.5 \times 25.0 \times 0.160^2 =$ **0.320 kg·m²**

Percentage difference = $(0.330 - 0.320)/0.320 \times 100 =$ **3.1 %**

Result

The moment of inertia of the flywheel determined experimentally is **0.330 kg·m²**, while its theoretical value is **0.320 kg·m²**. The difference is 3.1%, which is due to friction at the bearings and is within acceptable limits.

Precautions

- The thread must be wound uniformly on the axle and should not overlap.
- Start the stopwatch exactly when the weight is released and stop it when the weight touches the ground.
- Avoid parallax while measuring the height and radius.
- Take several trials and use the mean time to reduce random error.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 02

Study of different components of a Universal Testing Machine (UTM) and their functions

Aim

To study the construction and working of a Universal Testing Machine and to identify its different components with their functions.

Apparatus / Equipment Required

Universal Testing Machine (capacity 400 kN / 1000 kN) with accessories; Tension grips, compression platens, extensometer, dial gauge, measuring scale.

Theory

A Universal Testing Machine is used to conduct tension, compression, bending and shear tests on metals, timber and concrete. It consists of a load frame with a fixed crosshead and a movable crosshead, a hydraulic loading system and a load-measuring arrangement.

The load applied is measured either by a pendulum dynamometer or by a pressure gauge/load cell, and the deformation is measured by an extensometer or a dial gauge. From the load-deformation curve the yield strength, ultimate strength, modulus of elasticity and percentage elongation of the material can be determined.

Procedure

1. The machine was examined with the power supply switched off and the parts were identified.
2. The load frame, upper fixed crosshead and lower movable crosshead were studied.
3. The hydraulic pump, control valves, pressure gauge and load-indicating dial were identified.
4. The tension grips, compression platens, bending and shear attachments were examined.
5. The extensometer, dial gauge and the autographic recorder were studied along with their use.
6. The working of the machine — gripping, loading, rate control and unloading — was demonstrated.

Observation

S. No.	Component	Function
1	Base and columns	Support the frame and guide the movable crosshead
2	Fixed (upper) crosshead	Holds one end of the specimen through grips
3	Movable (lower) crosshead	Applies load to the specimen by hydraulic pressure
4	Hydraulic pump and valves	Generate and control the oil pressure for loading
5	Load measuring dial / load cell	Indicates the load applied on the specimen
6	Extensometer / dial gauge	Measures elongation or deformation of the specimen
7	Grips and platens	Hold the specimen for tension, compression, bending or shear tests
8	Autographic recorder	Plots the load-deformation curve automatically

Machine capacity: 400 kN; loading rate for steel test: 10 N/mm² per second.

Calculation

No calculation is required in this study. The relation between load and stress is:

Stress = Load / Cross-sectional area, and Strain = Change in length / Original length.

Modulus of elasticity E = Stress / Strain, obtained from the initial straight portion of the load-deformation curve.

Result

The different components of the Universal Testing Machine were studied and their functions were understood. The machine can perform tension, compression, bending and shear tests, and the load is measured by the pendulum dynamometer / load cell while deformation is measured by the extensometer.

Precautions

- Never operate the machine without the permission of the laboratory in charge.
- Ensure that the specimen is properly gripped and aligned before loading.
- Do not exceed the capacity of the machine.
- Keep hands away from the crossheads and grips during the test.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 03

Tension test on mild steel bar as per IS: 1608

Aim

To determine the yield strength, ultimate tensile strength, breaking strength and percentage elongation of a given mild steel bar by a tension test.

Apparatus / Equipment Required

Universal Testing Machine with tension grips; Mild steel specimen (12 mm diameter, gauge length 60 mm), punch, steel scale, vernier caliper.

Theory

When a ductile material such as mild steel is subjected to a gradually increasing axial tensile load it passes through the proportional limit, elastic limit, upper and lower yield points, strain hardening and ultimate load, and finally fractures. The yield stress is the load at which plastic deformation begins and the ultimate stress is the maximum load divided by the original area, while the percentage elongation of the gauge length after fracture indicates the ductility of the material.

Formulae Used

- Area, $A = \pi d^2 / 4$
- Yield stress = yield load / original area ; Ultimate stress = maximum load / original area
- Percentage elongation = [(L■ – L■) / L■] × 100
- Percentage reduction in area = [(A■ – A■) / A■] × 100

Procedure

1. The diameter of the specimen was measured at three places with a vernier caliper and the mean was taken.
2. A gauge length of 60 mm was marked on the specimen and the specimen was fitted in the grips of the UTM.
3. The load was applied gradually and the yield point, maximum load and breaking load were noted.
4. The load-deformation curve was recorded on the autographic recorder.
5. The two broken pieces were joined and the final gauge length was measured.
6. The stresses and percentage elongation were calculated.

Observation

Particular	Value	Unit
Original diameter / area	12.0 / 113.1	mm / mm²
Original gauge length, L■	60.0	mm
Yield load / maximum load	33.0 / 47.0	kN
Breaking load	36.5	kN
Final gauge length, L■	78.0	mm
Diameter at fracture	8.4	mm

Material: mild steel (Fe 415 grade); rate of loading 10 N/mm²/s. Readings are sample values.

Calculation

$A = \pi \times 12.0^2/4 =$ **113.1 mm²**

Yield stress = $33.0 \times 10^3 / 113.1 =$ **291.8 N/mm²**

Ultimate tensile stress = $47.0 \times 10^3 / 113.1 =$ **415.6 N/mm²**

Breaking stress = $36.5 \times 10^3 / 113.1 =$ **322.7 N/mm²**

Percentage elongation = (78.0 – 60.0)/60.0 × 100 = **30.0 %**; final area = 55.4 mm², hence reduction in area = (113.1 – 55.4)/113.1 × 100 = **51.0 %**

Result

For the given mild steel specimen, the yield stress is **291.8 N/mm²**, the ultimate tensile strength is **415.6 N/mm²**, the breaking stress is **322.7 N/mm²**, the percentage elongation is **30%** and the reduction in area is **51%**. The material is ductile and conforms to Fe 415 grade steel.

Precautions

- Measure the diameter accurately at three sections and use the mean value.
- Apply the load gradually and without shock, at the specified rate.
- Mark the gauge length before the test and measure the final length after joining the broken pieces.
- Note the yield point carefully — the pointer may pause or drop at the yield stage.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 04

Compression test on a mild steel specimen using Universal Testing Machine

Aim

To determine the compressive strength and behaviour of a given mild steel specimen under axial compression using a Universal Testing Machine.

Apparatus / Equipment Required

Universal Testing Machine with compression platens; Mild steel specimen (12 mm diameter, 24 mm length), vernier caliper, steel scale.

Theory

When a short specimen is subjected to axial compression it shortens and bulges laterally. For a ductile material such as mild steel, the specimen does not fracture but flattens, so the test gives the yield strength in compression rather than a breaking stress.

The compressive stress is the load divided by the original cross-sectional area. For ductile materials the compressive yield stress is nearly equal to the tensile yield stress, whereas brittle materials are much stronger in compression than in tension.

Formulae Used

- Area, $A = \pi d^2 / 4$
- Compressive stress = load / original area
- Percentage decrease in height = [(h■ – h■) / h■] × 100

Procedure

1. The diameter and height of the specimen were measured with a vernier caliper.
2. The specimen was placed centrally between the two compression platens of the UTM.
3. The load was applied gradually and the yield load was noted from the dial/autographic record.
4. Loading was continued until the specimen flattened to about half of its original height.
5. The final height was measured and the compressive stress was calculated.

Observation

Particular	Value	Unit
Original diameter	12.0	mm
Original height, h■	24.0	mm
Original area	113.1	mm²
Yield load (compression)	32.0	kN
Maximum load applied	58.0	kN
Final height, h■	14.0	mm

Material: mild steel; loading rate 10 N/mm²/s. Readings are sample values.

Calculation

$A = \pi \times 12^2/4 =$ **113.1 mm²**

Compressive yield stress = $32.0 \times 10^3 / 113.1 =$ **283.0 N/mm²**

Maximum compressive stress = $58.0 \times 10^3 / 113.1 =$ **512.8 N/mm²**

Percentage decrease in height = (24.0 – 14.0)/24.0 × 100 = **41.7 %**

The compressive yield stress (283 N/mm²) is nearly equal to the tensile yield stress of mild steel (about 292 N/mm²), which confirms the ductile behaviour of the material.

Result

The compressive yield stress of the given mild steel specimen is **283.0 N/mm²** and the specimen flattened by about 42% of its original height without fracture. The material behaves as a ductile material in compression.

Precautions

- Place the specimen exactly at the centre of the lower platen to avoid eccentric loading.
- Apply the load slowly and uniformly; avoid shock loading.
- Do not continue loading beyond the capacity of the machine or after heavy bulging.
- Measure the height and diameter before and after the test with the same instrument.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 05

Flexural test on a timber beam in both orientations as per IS: 2408

Aim

To determine the flexural strength (modulus of rupture) of a given timber beam by testing it in two orientations of the rectangular section.

Apparatus / Equipment Required

Universal Testing Machine / flexure testing arrangement with two supports and a loading nose; Timber specimen (50 mm × 75 mm × 750 mm), vernier caliper, steel scale, dial gauge.

Theory

When a simply supported beam is loaded at the centre, bending stresses are set up in the section. The maximum bending stress at failure, called the modulus of rupture, is given by the flexure formula f = M/Z, where M is the maximum bending moment and Z the section modulus.

For a rectangular section of width b and depth d, Z = bd²/6. The flexural strength therefore depends on the orientation of the section: a beam is much stronger when loaded on its narrower face, because the depth is then greater.

Formulae Used

- Maximum bending moment, M = W L / 4 (central point load)
- Section modulus, Z = b d² / 6
- Modulus of rupture, f = M / Z = 3 W L / (2 b d²)

Procedure

1. The dimensions of the timber specimen were measured accurately.
2. The specimen was placed on the two supports of the testing arrangement with a span of 600 mm, on its wider face (orientation 1).
3. The load was applied at the centre through the loading nose at a uniform rate.
4. The maximum load at failure was noted and the modulus of rupture calculated.
5. The test was repeated on an identical specimen placed on its narrower face (orientation 2).
6. The two values were compared.

Observation

Orientation	b (mm)	d (mm)	Span L (mm)	Failure load W (kN)
1 — on wider face	50	75	600	3.20
2 — on narrower face	75	50	600	1.42

Timber: teak; moisture content about 12%. Readings are sample values.

Calculation

Orientation 1: f = 3 W L/(2 b d²) = (3 × 3200 × 600)/(2 × 50 × 75²) = 5,760,000/562,500 = **10.24 N/mm²**

Orientation 2: f = (3 × 1420 × 600)/(2 × 75 × 50²) = 2,556,000/375,000 = **6.82 N/mm²**

Ratio of strengths = 10.24/6.82 = 1.50, which equals the ratio of the depths (75/50 = 1.5), showing that the flexural strength is proportional to the depth of the section.

Result

The modulus of rupture of the given timber is **10.24 N/mm²** when the beam is loaded on its wider face and **6.82 N/mm²** when loaded on its narrower face. The strength in the first orientation is 1.5 times that in the second, confirming that the flexural strength is proportional to the square of the depth.

Precautions

- The specimen must be free from knots, cracks and other visible defects.
- Apply the load exactly at the centre of the span and without impact.
- Measure the dimensions at the section of failure.
- Keep the supports and loading nose parallel and firmly in contact with the specimen.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 06

Shear strength test on a given specimen

Aim

To determine the ultimate shear strength of a given mild steel specimen by a single shear test.

Apparatus / Equipment Required

Universal Testing Machine with shear test attachment (shear block and punch); Mild steel specimen (12 mm diameter), vernier caliper, steel scale.

Theory

When a specimen is subjected to a force acting parallel to its cross-section, shear stress is developed. In a single shear test the specimen is held in a shear block and a punch shears it across one plane, so the resisting area is the cross-sectional area of the specimen.

The shear strength is the maximum load divided by the sheared area. For mild steel the ultimate shear strength is approximately 0.5 to 0.6 times the ultimate tensile strength.

Formulae Used

- Sheared area, $A = \pi d^2 / 4$
- Ultimate shear strength, τ = shear load / sheared area

Procedure

1. The diameter of the specimen was measured and its cross-sectional area calculated.
2. The specimen was inserted in the shear block of the attachment, which was placed in the UTM.
3. The load was applied gradually until the specimen was sheared off.
4. The maximum shear load was noted from the load indicating dial.
5. The shear strength was calculated and compared with the tensile strength of the material.

Observation

Particular	Value	Unit
Diameter of specimen	12.0	mm
Sheared area	113.1	mm²
Shear load at failure	24.0	kN
Type of shear	Single shear	—

Material: mild steel. Readings are sample values.

Calculation

$A = \pi \times 12^2/4 =$ **113.1 mm²**

$\tau = 24.0 \times 10^3 / 113.1 =$ **212.2 N/mm²**

Compare with tensile strength of mild steel (≈ 415 N/mm²): $\tau/f_u = 212.2/415.6 = 0.51$, which agrees with the usual relation $\tau \approx 0.5\ f_u$.

Result

The ultimate shear strength of the given mild steel specimen is **212.2 N/mm²**, which is about 51% of its ultimate tensile strength, as expected for ductile steel.

Precautions

- Place the specimen exactly in the centre of the shear block to avoid bending.
- Tighten the attachment firmly so that the specimen does not slip.
- Apply the load slowly and uniformly until shear occurs.
- Measure the diameter at the sheared section after the test.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 07

Izod impact test on a given specimen

Aim

To determine the impact strength (toughness) of a given metal specimen by the Izod impact test.

Apparatus / Equipment Required

Izod impact testing machine with a 167 J / 300 J pendulum hammer; Izod test specimen with a V-notch (10 mm × 10 mm section, 75 mm length), vernier caliper.

Theory

Toughness is the ability of a material to absorb energy before fracture. The Izod test measures this energy by breaking a notched specimen with a swinging pendulum; the specimen is held as a cantilever and struck at the notch.

The energy absorbed by the specimen is the difference between the energy of the hammer before and after fracture, and the impact strength is expressed as the energy absorbed per unit area at the notch.

Formulae Used

- Energy absorbed, E = W (h■ – h■) (J)
- Impact strength = energy absorbed / area of the notch (J/mm²)
- Impact value = energy absorbed by specimen / energy of the hammer × 100

Procedure

1. The dimensions of the specimen and the notch were measured.
2. The machine was checked with the pointer at the zero position and the pendulum was raised to its starting position.
3. The specimen was clamped in the vice with the notch facing the striker and the pointer was set.
4. The pendulum was released and the energy absorbed by the specimen was read from the scale.
5. The test was repeated on two more specimens and the mean impact strength was calculated.

Observation

Specimen	Energy absorbed (J)	Notch area (mm²)	Impact strength (J/mm²)
1	12.0	80	0.150
2	11.5	80	0.144
3	12.5	80	0.156
Mean	12.0	80	0.150

Material: mild steel; notch 45° V-notch, 2 mm deep. Readings are sample values.

Calculation

Mean energy absorbed = (12.0 + 11.5 + 12.5)/3 = **12.0 J**

Notch area = 10 mm × (2 × 4 mm) = 80 mm² (approximate fractured area)

Impact strength = 12.0 / 80 = **0.150 J/mm²**

Impact value = (12.0 / 167) × 100 = **7.2 %** of the hammer energy

Result

The Izod impact strength of the given mild steel specimen is **12.0 J** (0.150 J/mm²), which indicates good toughness and ductility of the material.

Precautions

- The pendulum must be locked in the raised position before the specimen is fixed.
- Keep clear of the swing path of the pendulum during the test.
- Locate the specimen accurately in the vice with the notch exactly opposite the striker.
- Do not touch the specimen or the machine before the pendulum comes to rest.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 08

Charpy impact test on a metal specimen

Aim

To determine the impact energy absorbed by a given metal specimen using the Charpy impact test.

Apparatus / Equipment Required

Charpy impact testing machine with pendulum hammer and scale; Charpy V-notch specimen (10 mm × 10 mm × 55 mm), vernier caliper, setting gauge.

Theory

In the Charpy test the notched specimen is supported as a simple beam at both ends and is struck by the pendulum at the centre, on the face opposite the notch. Because the specimen is supported at both ends, the Charpy test gives a slightly different measure of toughness than the Izod test.

The energy absorbed in breaking the specimen is read directly from the graduated scale of the machine and is a measure of the resistance of the material to sudden loading.

Formulae Used

- Energy absorbed, $E = W(\cos\beta - \cos\alpha)$ (J)
- Impact strength = energy absorbed / cross-sectional area at the notch (J/mm²)

Procedure

1. The specimen was prepared with a 45° V-notch of 2 mm depth and its dimensions were measured.
2. The pendulum was raised to the starting angle and the pointer was set to zero.
3. The specimen was placed centrally on the supports with the notch away from the striker.
4. The pendulum was released and the energy absorbed was read from the scale after the swing.
5. The test was repeated on two more specimens and the mean value was taken.

Observation

Specimen	Energy absorbed (J)	Area at notch (mm²)	Impact strength (J/mm²)
1	15.0	80	0.188
2	14.5	80	0.181
3	15.5	80	0.194
Mean	15.0	80	0.188

Material: mild steel; Charpy V-notch specimen. Readings are sample values.

Calculation

Mean energy absorbed = (15.0 + 14.5 + 15.5)/3 = **15.0 J**

Impact strength = 15.0 / 80 = **0.188 J/mm²**

The Charpy value (15.0 J) is slightly higher than the Izod value (12.0 J) of the same material because the specimen is supported at both ends and the test geometry is different.

Result

The Charpy impact energy absorbed by the given mild steel specimen is **15.0 J** (0.188 J/mm²), indicating good impact toughness of the material.

Precautions

- Place the specimen exactly on the supports using the setting gauge.
- Ensure that the notch is on the side away from the striker.
- Release the pendulum smoothly from the correct height.
- Read the scale after the pendulum has completed its forward swing.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 09

Brinell hardness test on a metal specimen

Aim

To determine the Brinell hardness number (BHN) of a given metal specimen.

Apparatus / Equipment Required

Brinell hardness testing machine with a hardened steel ball indenter (10 mm); Microscope or Brinell index measuring scale, specimen, standard test block.

Theory

Hardness is the resistance of a material to indentation. In the Brinell test a hardened steel ball of diameter D is pressed into the surface of the specimen under a standard load P and the diameter d of the resulting indentation is measured after removing the load.

The Brinell hardness number is the load divided by the curved surface area of the indentation, and is expressed in kg/mm² (or in MPa, numerically equal to BHN × 9.81).

Formulae Used

- $BHN = 2P / [\pi D (D - \sqrt{(D^2 - d^2)})]$
- where P = load in kg, D = diameter of ball in mm and d = diameter of indentation in mm
- Load $P = 30 D^2$ for steel (i.e. 3000 kg for a 10 mm ball)

Procedure

1. The surface of the specimen was ground smooth and cleaned.
2. The specimen was placed on the anvil of the machine and the 10 mm ball indenter was brought into contact with the surface.
3. A load of 3000 kg was applied and maintained for 15 seconds.
4. The load was released, the specimen was removed and the diameter of the indentation was measured in two perpendicular directions with the microscope.
5. The mean diameter was used to calculate the Brinell hardness number.

Observation

Test	Load P (kg)	Ball dia D (mm)	Indentation dia d (mm)	BHN
1	3000	10	4.20	207
2	3000	10	4.25	202
3	3000	10	4.15	212
Mean	3000	10	4.20	207

Material: mild steel; dwell time 15 s. Readings are sample values.

Calculation

For d = 4.20 mm: $\sqrt{(D^2 - d^2)} = \sqrt{(100 - 17.64)} = \sqrt{82.36} = 9.075$ mm

$BHN = 2 \times 3000 / [\pi \times 10 \times (10 - 9.075)] = 6000 / (31.416 \times 0.925) = 6000/29.06 =$ **206.5**

Mean BHN of the three tests = (207 + 202 + 212)/3 ≈ **207**

Result

The Brinell hardness number of the given mild steel specimen is **207** (≈ 2030 N/mm²).

Precautions

- The test surface must be clean, smooth and flat, and free from scale.
- Keep the indentation away from the edge of the specimen (at least 2.5 d).
- Apply the load slowly and maintain it for the specified dwell time.
- Measure the indentation diameter in two perpendicular directions and use the mean.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 10

Rockwell hardness test on a metal specimen

Aim

To determine the Rockwell hardness (HRC) of a given hardened steel specimen.

Apparatus / Equipment Required

Rockwell hardness testing machine with diamond cone indenter (120°); Specimen, standard test block, anvil.

Theory

In the Rockwell test the depth of indentation is measured directly by the machine instead of the diameter, which makes the test very quick. A minor load of 10 kg is first applied to seat the indenter, and then a major load of 150 kg is applied; the Rockwell hardness is read on the dial after the major load is removed.

The Rockwell C scale (HRC) uses a diamond cone indenter with a total load of 150 kg and is suitable for hardened steel. Higher hardness gives a higher HRC value, and each division of the dial corresponds to 0.002 mm of indentation depth.

Formulae Used

- HRC = 100 – (depth of indentation in divisions × 0.002 mm)
- The hardness is read directly on the dial gauge of the machine.

Procedure

1. The specimen surface was cleaned and polished, and the specimen was placed on the anvil.
2. The diamond cone indenter was brought into contact with the surface and the minor load of 10 kg was applied.
3. The dial was set to zero and the major load of 150 kg was applied for 10–15 seconds.
4. The major load was removed and the hardness was read on the C scale of the dial.
5. Five readings were taken at different points and the mean HRC was calculated.

Observation

Reading	1	2	3	4	5	Mean
HRC value	42	41	43	42	41	41.8

Material: hardened steel; indenter: diamond cone 120°; total load 150 kg (Scale C).

Calculation

Mean HRC = (42 + 41 + 43 + 42 + 41)/5 = **41.8**

Approximate tensile strength from HRC ≈ 3.5 × HRC = 3.5 × 41.8 ≈ **146 kg/mm²** (≈ 1435 N/mm²), which is typical of hardened and tempered steel.

Result

The Rockwell hardness of the given hardened steel specimen is **41.8 HRC**, which indicates that the material is hard and suitable for tools and machine parts.

Precautions

- Test on a smooth, clean surface; roughness gives erroneous readings.
- Keep successive indentations at least 3 mm apart and away from the edges.
- Apply the minor load first and set the dial to zero before applying the major load.
- Clean the indenter and anvil after every test.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 11

Preparation of models of different types of trusses

Aim

To prepare models of different types of trusses and to distinguish between them on the basis of their geometry and member arrangement.

Apparatus / Equipment Required

Balsa wood / match sticks / straw, glue, thread, pins; Drawing sheet, scale, protractor, cutting blade, sketch pens.

Theory

A truss is a framework of straight members connected at their ends by pin joints so that the members carry mainly axial forces. A truss is called perfect if the number of members m and the number of joints j satisfy $m = 2j - 3$; if $m < 2j - 3$ the truss is deficient (unstable), and if $m > 2j - 3$ it is redundant.

Common types of trusses are the king post truss (span up to 6 m), the queen post truss (6–9 m), the howe and pratt trusses, the warren truss, the fink truss and the north-light roof truss. The choice depends on the span, the load and the material used.

Formulae Used

- Degree of indeterminacy for a plane truss, $m = 2j - 3$ (perfect truss)
- where m = number of members and j = number of joints

Procedure

1. The required number of balsa wood pieces was cut to the calculated lengths of the members.
2. The joints were marked on a drawing sheet and the members were assembled as per the geometry of the truss.
3. The joints were glued and pinned and the model was allowed to dry.
4. Three models — king post, queen post and warren truss — were prepared in the same manner.
5. For every model, the number of members and joints were counted and the condition $m = 2j - 3$ was checked.
6. The models were labelled and compared with sketches.

Observation

Type of truss	Joints (j)	Members (m)	2j – 3	Remarks
King post	6	9	9	Perfect truss
Queen post	8	13	13	Perfect truss
Warren truss	10	17	17	Perfect truss
Deficient truss (without one diagonal)	6	8	9	Deficient — unstable

Models prepared to a scale of 1:50. Joints were pinned and glued.

Calculation

King post: $m = 2 \times 6 - 3 = 9$; actual members = 9 ✓ (perfect truss)

Queen post: $m = 2 \times 8 - 3 = 13$; actual members = 13 ✓ (perfect truss)

Warren: $m = 2 \times 10 - 3 = 17$; actual members = 17 ✓ (perfect truss)

Model with one member removed: $m = 8 < 9$, hence the truss becomes deficient and can collapse under load.

Result

Models of the king post, queen post and warren trusses were prepared successfully, and all of them satisfy the condition $m = 2j - 3$ for a perfect truss. The model with one member removed was found to be deficient, confirming the need for the correct number of members.

Precautions

- Cut the members to accurate lengths, otherwise the joints will not close properly.
- Keep the joints exactly at the marked points to maintain the geometry of the truss.
- Allow sufficient time for the adhesive to set before handling the model.
- Handle the models carefully while shifting and displaying them.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 12

Shear force and bending moment diagrams for a simply supported beam under point and distributed loads

Aim

To draw the shear force and bending moment diagrams for a simply supported beam carrying a point load and a uniformly distributed load, and to locate the point of maximum bending moment.

Apparatus / Equipment Required

Drawing sheet, scale, pencil, French curves, calculator; Data of the beam: span, loads and positions.

Theory

Shear force at a section is the algebraic sum of all vertical forces on one side of the section, and bending moment is the algebraic sum of the moments of those forces about the section. The relation between load, shear force and bending moment is dM/dx = V and dV/dx = w.

For a simply supported beam the bending moment is zero at both ends, and the maximum bending moment occurs where the shear force changes sign (point of zero shear).

Formulae Used

- Reactions: $\Sigma M_A = 0$ and $\Sigma V = 0$
- Shear force V_x = sum of upward forces on the left of the section
- Bending moment $M_x = \Sigma$ (force × lever arm)
- Maximum BM occurs at the section where V = 0

Procedure

1. The beam of span 6 m carrying a point load of 10 kN at 2 m from the left support and a UDL of 2 kN/m over the whole span was considered.
2. The reactions at the supports were calculated using the equations of equilibrium.
3. Shear force was calculated at every critical section (supports, under the point load and points of zero shear).
4. Bending moment was calculated at the same sections.
5. The shear force and bending moment diagrams were drawn to a suitable scale below the beam sketch.

Observation

Section (m from A)	Shear force (kN)	Bending moment (kNm)	Remarks
0 (A)	+12.67	0	Left support
2.0 (just left of load)	+8.67	+25.33	Under 10 kN load
2.0 (just right of load)	–1.33	+25.33	Sign change
2.67	0	+25.79	Maximum BM (V = 0)
6.0 (B)	–9.33	0	Right support

Span 6 m; point load 10 kN at 2 m from A; UDL 2 kN/m over the whole span.

Calculation

Taking moments about A: $R_B \times 6 = 10 \times 2 + (2 \times 6) \times 3 = 20 + 36 = 56 \rightarrow R_B =$ **9.33 kN**

R_A = total load – R_B = (10 + 12) – 9.33 = **12.67 kN**

SF at A = +12.67 kN; just left of the load = 12.67 – 2 × 2 = +8.67 kN; just right = 8.67 – 10 = –1.33 kN

Point of zero shear: 12.67 – 2x – 10 = 0 → x = **1.33 m** from the load, i.e. 2.67 m from A

M at 2 m = 12.67 × 2 – 2 × 2²/2 = 25.33 – 4.00 = **21.33 kNm** (left of load)

M_{max} at x = 2.67 m = 12.67 × 2.67 – 2 × 2.67²/2 – 10 × 0.67 = 33.83 – 7.13 – 6.70 = **20.00 kNm**

Check: M at B = 12.67 × 6 – 10 × 4 – 2 × 6²/2 = 76.02 – 40 – 36 = 0.02 ≈ **0** ✓

Result

The reactions are R_A = 12.67 kN and R_B = 9.33 kN. The maximum bending moment is **20.00 kNm** at 2.67 m from the left support, where the shear force changes sign, and the bending moment is zero at both ends.

Precautions

- Follow a proper sign convention throughout (upward force = positive shear).
- Calculate the reactions correctly before drawing the diagrams.
- Mark the points of zero shear and maximum bending moment clearly.
- Draw the diagrams to a stated scale and label the values at all critical sections.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 13

Determination of principal stresses and their plane using Mohr's circle

Aim

To determine the principal stresses, maximum shear stress and the position of the principal planes for a given state of stress using Mohr's circle.

Apparatus / Equipment Required

Drawing sheet, compass, protractor, scale, pencil, calculator; Given state of stress: σ_x = 80 N/mm², σ_y = 20 N/mm², τ = 30 N/mm².

Theory

At a point in a stressed body there are always two mutually perpendicular planes on which the shear stress is zero; the normal stresses on these planes are called principal stresses. Their values can be found analytically or graphically by Mohr's circle.

In Mohr's circle the normal stresses are plotted on the horizontal axis and the shear stresses on the vertical axis. The circle is drawn with the centre at $((\sigma_x + \sigma_y)/2, 0)$ and radius equal to the maximum shear stress; the points where the circle cuts the horizontal axis give the principal stresses.

Formulae Used

- Centre of circle, $C = (\sigma_x + \sigma_y) / 2$
- Radius, $R = \sqrt{[\,((\sigma_x - \sigma_y)/2)^2 + \tau^2\,]}$
- $\sigma_1 = C + R$; $\sigma_2 = C - R$; $\tau_{max} = R$
- $\tan 2\theta = 2\tau / (\sigma_x - \sigma_y)$

Procedure

1. The given values of σ_x, σ_y and τ were noted.
2. The centre C and radius R of Mohr's circle were calculated.
3. The circle was drawn to a suitable scale with C as centre on the σ-axis.
4. The points of intersection of the circle with the σ-axis were marked as σ_1 and σ_2.
5. The maximum shear stress was measured as the radius and the angle 2θ was measured from the circle.
6. The values were compared with those obtained analytically.

Observation

Quantity	Graphical value	Analytical value	Unit
Centre of circle	50.0	50.0	N/mm²
Radius (τ_{max})	42.4	42.43	N/mm²
Major principal stress σ_1	92.4	92.43	N/mm²
Minor principal stress σ_2	7.6	7.57	N/mm²
Angle of principal plane θ	22.5°	22.5°	degree

Scale: 1 cm = 10 N/mm². σ_x = 80, σ_y = 20, τ = 30 N/mm².

Calculation

$C = (80 + 20)/2 =$ **50 N/mm²**

$R = \sqrt{[\,((80 - 20)/2)^2 + 30^2\,]} = \sqrt{(900 + 900)} = \sqrt{1800} =$ **42.43 N/mm²**

$\sigma_1 = 50 + 42.43 =$ **92.43 N/mm²**; $\sigma_2 = 50 - 42.43 =$ **7.57 N/mm²**

$\tau_{max} = R =$ **42.43 N/mm²**

$\tan 2\theta = (2 \times 30)/(80 - 20) = 60/60 = 1 \rightarrow 2\theta = 45° \rightarrow \theta =$ **22.5°**

Result

For the given state of stress the principal stresses are σ_1 **= 92.43 N/mm²** and σ_2 **= 7.57 N/mm²**, the maximum shear stress is **42.43 N/mm²**, and the principal planes are inclined at **22.5°** to the plane of σ_x. The graphical and analytical values agree closely.

Precautions

- Choose a suitable scale so that the circle fits on the sheet.
- Measure the angles from the correct reference axis (2θ on Mohr's circle).
- Plot the shear stress with the correct sign convention.
- Verify the graphical results with the analytical formulae.

Date: _____ Teacher's Signature: _____

References and Useful Standards

Practical file likhne ke liye nimnlikhit textbooks, IS codes aur curriculum outcomes ka sahayata liya gaya hai. Viva mein IS code number poochhe jaate hain, isliye inhe ek baar zaroor padh lein.

S. No.	Reference / Standard
1	R.K. Bansal, *Strength of Materials*, Laxmi Publications.
2	B.C. Punmia, *Strength of Materials and Mechanics of Structures*, Laxmi Publications.
3	S. Ramamrutham, *Strength of Materials*, Dhanpat Rai Publishing Co.
4	IS: 1608 — tensile testing; IS: 2408 — timber; IS: 1977 — Izod impact.
5	IS: 1757 — Charpy impact; IS: 2281 — Brinell hardness; IS: 1586 (Part 2) — Rockwell hardness.
6	SBTE Bihar curriculum for Strength of Material (2415303 / P2415303).

Note: Is file mein di gayi observation readings sample hain — samajhne aur format ke liye. Apne practical class mein actual readings le kar observation table dobara bharein aur result calculate karein. Practical ki sankhya apne college ke lab manual ke anusaar ho sakti hai.